

Course Description

Title of Course: Data Analysis and Visualization Lab

Course Code: CS223

L-T-P scheme: 0-0-1

Credit: 1

Course Coordinator: Dr. Amit Kumar

Prerequisite: Python programming with basic mathematical knowledge.

Objective:

This course will introduce the main concepts of visual data analytics with a hands-on tutorial using Python. Further, it aims at learning about how to create effective charts and interactive dashboards will provide the student a very useful skill applicable in many business scenarios.

Learning Outcomes:

At the end of the course, students will be able to:

Data Analysis and Visualization Lab		
Course Outcome	Description	BCL
CO1	Understand and describe the main concepts of data visualization using MS excel.	3
CO2	Demonstrate the learning of numerical computation library and its functions in python programming.	3
CO3	Demonstrate the learning of data visualization using in python programming.	3
CO4	Create plots to show insights in the data using python programming.	4
CO5	Design python code to Create ad-hoc reports, data visualization on large datasets.	4

Course Content:

Unit 1: Environment Setup and Excel Data Import - Setting up Python environment (Anaconda, Jupyter Notebook, VS Code), Installing and managing essential libraries (pandas, NumPy, Matplotlib, Seaborn, openpyxl), Reading Excel files using pandas (read_excel, ExcelFile, sheet navigation), Working with advanced Excel operations (workbook manipulation, cell operations), Excel file formats and compatibility considerations (xlsx, xls, xlsxm).

Unit 2: Basic Plots using matplotlib - Line Plot, Bar Chart, Horizontal Bar Chart, Scatter Plot, Pie Chart, Histogram, Box Plot, Area Plot, Stacked Bar Chart, Bubble Chart on random data set and specific data sets.

Unit 3: Statistical Visualization with Seaborn - Distribution Plot, Violin Plot, Heatmap, Joint Plot, Swarm Plot, Strip Plot, KDE Plot, Regression Plot on the tips dataset, Pair Plot on the iris dataset., Count Plot on titanic data set.

Unit 4: Multi-panel, 3D & Specialized Plots - Subplots (2x2 Grid), 3D Surface Plot, 3D Scatter Plot, Polar Plot, Contour Plot, Error Bar Plot, Logarithmic Scale Plot, Twin Axes Plot, Annotated Plot, Custom Styled Plot,

Unit 5: Time Series, Text, Networks & Hierarchical Data- Time Series Line Plot, Heatmap Calendar (Flights Pivot) on flights data set, Rolling Average Plot, Word Cloud, Network Graph, Treemap (using Plotly), Sankey Diagram (using Plotly), Radar (Spider) Chart, Candlestick Chart, Waterfall Chart.

Unit 6: Interactive Plotly Charts & Real Dataset EDA- Interactive Scatter Plot, Interactive Line Chart, Interactive Bar Chart, Choropleth Map, 3D Scatter with Plotly, Animated Bubble Chart, Full EDA on Titanic Dataset, Full EDA on Penguins Dataset, Custom Color Palettes & Themes, Exporting & Saving Visualizations.

Teaching Methodology:

This course is introduced to help students understand the discipline of data visualization. The programming tool used to teach this course is Python. Starting from the basic mathematical tools, the student will slowly be exposed to inferential statistics, and later to data analysis and visualize the data in different ways.

Evaluation Scheme:

Evaluations		Marks	Remarks
P-1		15 Marks	
P-2		15 Marks	
Continuous Evaluations	Viva	20 Marks	
	Demonstration	20 Marks	
	Lab Record	15 Marks	
	Discipline and Punctuality and Attendance	15 Marks	
Total		100 Marks	

Learning Resources:

Lab exercises are uploaded on the server (added from time to time).

Reference Material:

- “Data Visualization with Python and JavaScript Scrape, Clean, Explore & Transform Your Data” by Kyran Dale
- “Beginning Python Visualization Crafting Visual Transformation Scripts” by Shai Vaingast
- “Data Visualization in Python” by Daniel Nelson
- “Python for Data Analysis” by Wes McKinney.
- “The Data Visualization Workshop” by Mario Döbler and Tim Großmann

Online Resources:

1. <https://www.tableau.com/trial/data-analysis-software>
2. <https://aws.amazon.com/big-data/datalakes-and-analytics>
3. <https://www.coursera.org/in/articles/data-analysis-tools>
4. <https://www.kaggle.com/learn>
5. <https://nptel.ac.in/courses/106107220>
6. <https://nptel.ac.in/courses/102105556>